

Lesson 21: Sine & Cosine Rules — Practice Worksheet

Name: _____ Date: _____

Attempt every question. Show your working. The sections increase in difficulty.

Section 1 — Warm-up (foundation)

1. Which rule needs an angle and its opposite side?
2. Which rule is used for SAS (two sides + included angle)?
3. Write the area formula for a triangle (two sides + angle).
4. State the sine rule.
5. State the cosine rule for a^2 .
6. Sum of the angles in a triangle?

Section 2 — Core practice

7. Sine rule: $A = 50^\circ$, $a = 10$, $B = 30^\circ$. Find b (3 s.f.).
8. Sine rule: $A = 70^\circ$, $a = 12$, $C = 40^\circ$. Find c (3 s.f.).
9. Cosine rule: $b = 5$, $c = 6$, $A = 80^\circ$. Find a (3 s.f.).
10. Area: $a = 9$, $b = 10$, $C = 30^\circ$.
11. Area: $a = 7$, $b = 8$, $C = 50^\circ$.

12. Cosine rule: $b = 4$, $c = 7$, $A = 60^\circ$. Find a .

Section 3 — Challenge & exam-style

13. Triangle ABC: $a = 11$, $b = 8$, $C = 35^\circ$. Find c and the area.

14. Triangle: $a = 13$, $b = 14$, $c = 15$. Find angle A (3 s.f.).

15. Sine rule: $A = 45^\circ$, $a = 14$, $B = 60^\circ$. Find b .

16. A triangle has sides 8 and 11 with included angle 40° . Find the area and the third side.